ECEN 489 – Progress Report 2

Large Scale Artificial Life Simulation Using CUDA

Jonathan Gao

Texas A&M University

jobogao@tamu.edu

Accomplishments
Since last report, the project has gone from a basic CUDA prototype into a small but working artificial life simulation. It now includes GPU-based agents, food particles, movement, energy tracking, food detection and consumption, respawning, agent death, and a live Python visualization.

The agent system was redesigned using a structure of arrays layout to store position, energy, direction, and alive status, improving both simplicity and performance.

Agent behavior was added so agents search for nearby food and move toward it, or wander with a directional bias if none is found. Food can be consumed and respawned, and agents lose energy over time and die when it reaches zero.

Visualization also improved from a basic ASCII view to a real-time Python pipeline that reads directly from the simulation output. It displays agents, food, and status information without using intermediate files.

Several issues were resolved along the way, including repetitive movement patterns, agents getting stuck at boundaries, and simplifying communication between simulation and visualization.

Individual Contribution
I implemented the main CUDA simulation loop, including agent updates, food interaction, energy tracking, and death logic. I also built the Python visualization pipeline for real-time display of simulation data.

Remaining Work
The current simulation works at small scale but needs to grow while staying efficient and correct.

The food search is currently inefficient since each agent checks all food particles. This will be replaced with a spatial structure to limit searches to nearby food. There is also a race condition when multiple agents try to consume the same food, which will need an atomic or similar solution.

The simulation also lacks higher-level behavior such as reproduction, mutation, and population control. Visualization can be improved with better statistics and controls.

Key challenges include handling race conditions, keeping behavior deterministic, and scaling without making the system too complex.

Next Steps
The next phase will focus on correctness, basic life-cycle features, and scaling.

- Clean and document the current code
- Fix food consumption conflicts
- Add simple statistics (agent count, energy, food count)
- Begin scaling experiments

- Prepare a final demo showing full simulation behavior